

Akash Marsalin

Ph.D. Research Scholar | Nanomaterials Researcher | Cancer Theranostics | Magnetic Mesocrystals

Department of Physics and Nanotechnology
SRM Institute of Science and Technology (SRMIST)
Chengalpattu, Tamil Nadu, India

Welcome

Welcome to my personal academic website!

I am a Ph.D. Research Scholar at the **Department of Physics and Nanotechnology, SRM Institute of Science and Technology (SRMIST)** and a member of the **Key Laboratory of Emergent Materials (KLEM)**. My research focuses on designing multifunctional magnetic mesocrystals for advanced biomedical applications, particularly cancer theranostics.

By integrating magnetic hyperthermia, photothermal therapy, targeted drug delivery, and fluorescence bioimaging into a single nanoplatform, my goal is to develop innovative nanomaterials that contribute to safer, more effective, and personalized cancer treatment.

Thank you for visiting my website. I hope it provides insight into my research, academic journey, and professional interests.

Research Interests

- Magnetic Mesocrystals
 - Cancer Theranostics
 - Magnetic Hyperthermia
 - Photothermal Therapy
 - Drug Delivery
 - Biomedical Nanotechnology
 - Rare-Earth Luminescent Nanomaterials
 - Nanothermometry
 - Magnetic Nanoparticles
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Quick Highlights

 Ph.D. Research Scholar at SRMIST

 Member of Key Laboratory of Emergent Materials (KLEM)

 Published research and review articles in international journals

 Presented research in France and India

 Recipient of ANRF International Travel Support

 Best Oral Presentation Award

ABOUT ME

Hello! I'm **Akash Marsalin**, a Ph.D. Research Scholar in the **Department of Physics and Nanotechnology** at **SRM Institute of Science and Technology (SRMIST), India**, where I am pursuing research under the guidance of **Dr. Rajaboopathi Mani** as a member of the **Key Laboratory of Emergent Materials (KLEM)**.

My research lies at the intersection of **materials science, nanotechnology, and biomedicine**, with a particular focus on developing **multifunctional magnetic mesocrystals** for cancer theranostics. I am fascinated by the possibility of designing intelligent nanomaterials that can simultaneously diagnose and treat diseases. My current work explores magnetic hyperthermia, photothermal therapy, targeted drug delivery, fluorescence bioimaging, and nanothermometry using engineered Fe_3O_4 mesocrystals.

My interest in research began during my master's studies in Physics at **Periyar University**, where I investigated zinc-substituted cobalt ferrite nanoparticles for magnetic hyperthermia applications. This experience sparked my curiosity about magnetic nanomaterials and inspired me to pursue doctoral research in biomedical nanotechnology. During my academic journey, I also completed a research internship at the **National Centre for Nanoscience and Nanotechnology, University of Madras**, where I worked on lead-free polymer nanocomposites for X-ray shielding applications.

Since joining KLEM in 2023, I have had the opportunity to collaborate on exciting research projects, publish research and review articles in internationally recognized journals, and present my work at national and international conferences. Presenting my research at **Sorbonne University, Paris, France**, and receiving the **ANRF International Travel Support** and **Best Oral Presentation Award** have been important milestones in my doctoral journey.

Outside the laboratory, I believe that maintaining a balance between research and personal interests is essential for creativity and productivity. I represented both my **college and club handball teams** for nearly **four years**, competing at the **state level**, and I currently enjoy playing **badminton**. Sports have taught me discipline, resilience, teamwork, and perseverance—qualities that I apply in both research and everyday life.

Photography is another passion of mine. For more than **six years**, I have been involved in **photography, videography, and live media production** at **Jesus With Us Church, Karungal, Tamil Nadu, India**. Serving as a **Group Leader** for three years further strengthened my leadership, communication, event management, and organizational skills. These experiences have taught me how to work effectively under pressure while managing multiple responsibilities simultaneously.

Beyond academics, I enjoy **watching anime**, travelling, exploring emerging technologies, and continuously learning new skills. I appreciate anime for its creativity, storytelling, and themes of perseverance and self-improvement, which often inspire my own approach to research and personal growth.

Looking ahead, my goal is to establish an independent research career dedicated to developing multifunctional nanomaterials for precision medicine. I aspire to contribute to the advancement of cancer theranostics through interdisciplinary research while mentoring future scientists and fostering meaningful collaborations across the global scientific community.

RESEARCH

Research Overview

Cancer remains one of the world's leading health challenges, creating an urgent need for therapeutic approaches that are more effective, minimally invasive, and capable of targeting tumors with high precision. Nanotechnology has emerged as a promising platform to address these challenges by enabling multifunctional materials that combine diagnosis and therapy within a single system.

My doctoral research focuses on the design, synthesis, and application of **multifunctional magnetic mesocrystals** for cancer theranostics. These hierarchically assembled nanostructures exhibit unique structural, magnetic, and optical properties that make them attractive candidates for biomedical applications.

A major objective of my research is to engineer Fe₃O₄-based mesocrystals capable of integrating **magnetic hyperthermia**, **photothermal therapy**, **targeted drug delivery**, and **fluorescence bioimaging** into a unified therapeutic platform. By tailoring crystal growth, morphology, rare-earth doping, and surface functionality, I aim to improve therapeutic efficiency while enabling simultaneous imaging and treatment.

My research combines advanced nanomaterial synthesis, structural characterization, magnetic measurements, optical studies, and biological evaluation to understand how material properties influence biomedical performance. Through this interdisciplinary approach, I hope to contribute to the development of next-generation nanomaterials that support precision medicine and improve clinical outcomes.

RESEARCH AREAS

Magnetic Mesocrystals

Design and synthesis of hierarchically assembled magnetic mesocrystals through non-classical crystallization for biomedical applications.

Magnetic Hyperthermia

Development of highly efficient magnetic nanomaterials capable of generating localized heat under alternating magnetic fields for cancer treatment.

Photothermal Therapy

Engineering multifunctional nanoplateforms that efficiently convert light into heat for minimally invasive cancer therapy.

Cancer Theranostics

Combining therapeutic and diagnostic functionalities into a single nanomaterial platform for personalized medicine.

Drug Delivery

Development of targeted drug delivery systems capable of controlled release and improved therapeutic efficiency.

Luminescent Nanomaterials

Design of rare-earth-doped magnetic nanomaterials for fluorescence imaging and multimodal biomedical applications.

Nanothermometry

Developing luminescent probes capable of monitoring local temperature during therapeutic procedures.

Biomedical Nanotechnology

Applying nanotechnology to improve disease diagnosis, treatment, and healthcare through multifunctional materials.

RESEARCH HIGHLIGHTS

Throughout my doctoral research, I have focused on developing multifunctional magnetic mesocrystals with enhanced magnetic and optical properties for biomedical applications. My work has contributed to advancing nanomaterial design for cancer theranostics through interdisciplinary research that combines materials science, nanotechnology, and biomedicine.

Key Highlights

- Developed multifunctional Fe₃O₄ mesocrystals for magnetic hyperthermia and targeted drug delivery.
- Investigated rare-earth-doped magnetic mesocrystals for magneto-optical cancer therapy and fluorescence bioimaging.
- Published research and review articles in internationally recognized journals.
- Presented research at prestigious national and international conferences, including **Sorbonne University, Paris, France, IIT Guwahati, SRMIST, NIT Warangal, and VIT-AP University.**
- Awarded the **ANRF International Travel Support** to present research internationally.
- Received the **Best Oral Presentation Award** for research on magneto-optic synergistic cancer therapy.
- Working toward integrating magnetic hyperthermia, photothermal therapy, drug delivery, and imaging into a single multifunctional nanoplatform.

PUBLICATIONS

My research focuses on the development of multifunctional magnetic mesocrystals and advanced nanomaterials for biomedical applications, particularly cancer theranostics. My work spans experimental nanomaterial synthesis, structural characterization, magnetic property optimization, and multifunctional therapeutic applications. Through my publications, I aim to contribute to the growing field of nanomedicine by designing innovative materials that integrate diagnosis and therapy into a single platform.

Journal Articles

1. Multifunctional Fe₃O₄ Mesocrystals for Cancer Therapy: Integrating Hyperthermia and Targeted Drug Delivery

Journal of Materials Chemistry B (2025)

This research presents the development of multifunctional Fe₃O₄ mesocrystals with enhanced magnetic properties for combined magnetic hyperthermia and targeted drug delivery. The study demonstrates how controlled mesocrystal engineering can improve drug-loading efficiency, magnetic heating performance, and therapeutic potential, making these nanostructures promising candidates for next-generation cancer theranostics.

Research Highlights

- Development of highly ordered Fe₃O₄ mesocrystals.
- Enhanced magnetic hyperthermia efficiency.
- Controlled drug loading and release.
- Potential platform for multifunctional cancer therapy.

Keywords: Fe₃O₄ Mesocrystals • Magnetic Hyperthermia • Drug Delivery • Cancer
Theranostics • Nanomedicine

2. From Formation to Functionality: Insights into Mesocrystal Development and Applications

Journal of Solid State Chemistry, Volume 347 (2025), 125331

This comprehensive review discusses the evolution of mesocrystal research, covering non-classical crystallization mechanisms, synthesis strategies, characterization techniques, and applications in catalysis, sensing, energy storage, environmental remediation, and biomedicine. The article highlights recent advances and future opportunities in mesocrystal engineering, serving as a valuable reference for researchers entering this rapidly growing field.

Research Highlights

- Comprehensive review of mesocrystal formation mechanisms.
- Comparison of synthesis approaches.
- Biomedical and energy-related applications.
- Future perspectives for multifunctional mesocrystals.

Keywords: Mesocrystals • Non-Classical Crystallization • Nanomaterials • Biomedical Applications • Review

Conference Proceedings

Investigation of Structural and Magnetic Property of Fe₃O₄ and CoFe₂O₄ Nanoclusters for Biomedical Application

Springer Nature Conference Proceedings

This conference paper investigates the synthesis, structural characterization, and magnetic behavior of Fe₃O₄ and CoFe₂O₄ nanoclusters for biomedical applications. The work explores how composition and crystal structure influence magnetic properties, providing valuable insights into designing magnetic nanomaterials for hyperthermia and related biomedical technologies.

CONFERENCE PRESENTATIONS

Presenting research at national and international conferences has been an essential part of my doctoral journey. These events have provided opportunities to share my research findings, interact with experts from around the world, receive valuable feedback, and establish collaborations that broaden the impact of my work.

2026

15th International Conference on the Scientific and Clinical Applications of Magnetic Carriers

Venue: Sorbonne University, Paris, France

Organizer: Magnetic Microsphere Society

Date: 26–29 May 2026

Presentation Type: Oral Presentation

Presentation Title

Structural Engineering of Fe₃O₄ Mesocrystals for Enhanced Magneto-Optical Cancer Therapy

This work focused on engineering Fe₃O₄ mesocrystals with optimized structural and magnetic properties to improve magneto-optical cancer therapy. The presentation highlighted strategies to integrate magnetic hyperthermia with optical functionality, enabling more efficient and targeted cancer treatment. The conference provided an excellent platform to exchange ideas with leading researchers working in magnetic carriers and biomedical nanotechnology.

ANRF International Travel Support

To present this work internationally, I was awarded the **Conference Travel Grant–2026 (ITS/2026/001597)** by the **Anusandhan National Research Foundation (ANRF)** under the International Travel Support (ITS) scheme. This prestigious grant recognized the quality and international relevance of my research.

2nd International Conference on Advanced Functional Materials and Devices (AFMD-2026)

Organizer: Nanotechnology Research Centre (NRC), SRM Institute of Science and Technology

Venue: Chennai, India

Date: 19–24 March 2026

Presentation Type: Oral Presentation

Presentation Title

Eu³⁺-Doped Fe₃O₄ Mesocrystals for Magneto–Optic Synergistic Cancer Therapy and Bioimaging

This presentation demonstrated the successful integration of luminescent Eu³⁺ ions into Fe₃O₄ mesocrystals to achieve combined magnetic hyperthermia, photothermal therapy, and fluorescence imaging. The work highlighted the potential of multifunctional magnetic nanomaterials for simultaneous cancer diagnosis and treatment.

2025

International Workshop on Smart Materials for Biomedical Applications (SMARTBIO-2025)

Organizer: School of Advanced Sciences, VIT-AP University

Venue: Amaravati, Andhra Pradesh, India

Date: 16–20 December 2025

Presentation Type: Oral Presentation

Presentation Title

Magneto-Optic Thermal Heating with Eu-Doped Fe₃O₄ Mesocrystals for Synergistic Cancer Therapy and Bioimaging

This work demonstrated how Eu-doped Fe₃O₄ mesocrystals can simultaneously generate heat under magnetic fields while providing fluorescence imaging capability. The presentation received the **Best Oral Presentation Award**, recognizing the novelty and scientific significance of the research.

9th International Conference on Advanced Nanomaterials and Nanotechnology (ICANN 2025)

Organizer: Centre for Nanotechnology, IIT Guwahati

Venue: Guwahati, Assam, India

Date: 12–14 December 2025

Presentation Type: Oral Presentation

Presentation Title

Achieving Synergistic Cancer Therapy with Magnetic Hyperthermia and Luminescent Photothermal Agent: A Study of Eu-Doped Fe₃O₄ Mesocrystals

This presentation explored the integration of magnetic hyperthermia and photothermal therapy using Eu-doped Fe₃O₄ mesocrystals, emphasizing their multifunctional capabilities for synergistic cancer treatment and imaging.

8th International Conference on Nanoscience and Nanotechnology (ICONN 2025)

Organizer: SRM Institute of Science and Technology

Venue: Chennai, India

Date: 24–26 March 2025

Presented recent progress on multifunctional magnetic mesocrystals for biomedical applications and interacted with researchers working across nanoscience, advanced materials, and biomedical engineering.

2024

International Workshop on Recent Advancements in Magnetism and Magnetic Materials (iReAd MAGMA-2024)

Organizer: VIT-AP University

Venue: Amaravati, Andhra Pradesh, India

Date: 26–30 November 2024

Presentation Type: Oral Presentation

Presented ongoing doctoral research on magnetic mesocrystals for biomedical applications, discussing recent advances in magnetic hyperthermia and multifunctional nanomaterials.

International Conference on Emerging Multifunctional Materials and Devices for Sustainable Technologies (IEMDST-2024)

Organizer: National Institute of Technology Warangal

Venue: Warangal, Telangana, India

Date: 4–5 July 2024

Presentation Type: Oral Presentation

Presentation Title

Investigation of Structural and Magnetic Property of Fe_3O_4 and CoFe_2O_4 Nanoclusters for Biomedical Application

This presentation discussed the synthesis and characterization of magnetic nanoclusters and their potential use in biomedical applications, particularly magnetic hyperthermia. The conference proceedings were subsequently accepted for publication by **Springer Nature**, marking an important milestone in my research dissemination.

WORKSHOPS & PROFESSIONAL TRAINING

Continuous learning and professional development are integral to my research journey. I actively participate in workshops, hands-on training programs, and faculty development initiatives to strengthen both my technical expertise and research skills.

International Workshop on Smart Materials for Biomedical Applications (SMARTBIO-2025)

VIT-AP University | 16–20 December 2025

Participated in an international workshop focused on advanced smart materials for biomedical applications. Presented my research and received the **Best Oral Presentation Award**.

IEEE Around-the-Clock Around-the-Globe (AtC-AtG) Magnetism Workshop

18 November 2025

Participated in an international workshop covering recent developments in magnetism, magnetic materials, and their emerging technological applications.

Transforming Preclinical Science: Emerging Alternatives to Animal Testing

SRM Institute of Science and Technology | 22–23 August 2025

Attended a workshop discussing modern alternatives to conventional animal testing, ethical research practices, and emerging approaches in biomedical science.

Scientific Writing Workshop

ACS Publications & MIT World Peace University | 9 May 2025

Participated in a workshop on scientific writing, manuscript preparation, peer review, and strategies for publishing high-quality research in international journals.

Hands-on Training on Rietveld Refinement of X-Ray Diffraction Data

Centre for Advanced Computational Research | 25–31 March 2025

Completed intensive training on Rietveld refinement using GSAS-II, gaining practical experience in crystal structure refinement, phase analysis, and quantitative XRD data interpretation.

Computational Chemistry Training – Amsterdam Modeling Suite

University of Calicut | 27–28 February 2025

Participated in computational chemistry training focused on density functional theory (DFT), molecular simulations, and the Amsterdam Modeling Suite for materials research.

Research Methodology and Publication Ethics Workshops (2023)

During the initial phase of my doctoral research, I participated in workshops on research methodology, academic writing, publication ethics, and advanced characterization techniques.

These programs provided a strong foundation in responsible research practices, scientific communication, and experimental planning, which continue to support my doctoral research.

Professional Development

I strongly believe that research extends beyond the laboratory. Participating in conferences and workshops has enhanced my technical knowledge, scientific communication skills, and professional network. These experiences continue to inspire me to pursue interdisciplinary collaborations and contribute meaningfully to the global scientific community.

AWARDS & FELLOWSHIPS

Throughout my academic journey, I have been fortunate to receive recognition for my research contributions, scientific presentations, and academic excellence. These awards and fellowships have encouraged me to pursue innovative research in multifunctional nanomaterials and strengthen my commitment to advancing cancer theranostics.

ANRF International Travel Support (2026)

Anusandhan National Research Foundation (ANRF), India

Awarded the **International Travel Support (ITS/2026/001597)** to present my research at the **15th International Conference on the Scientific and Clinical Applications of Magnetic Carriers**, held at **Sorbonne University, Paris, France**. This competitive fellowship supported my participation in one of the leading international conferences in magnetic carriers and biomedical nanotechnology.

Best Oral Presentation Award (2025)

International Workshop on Smart Materials for Biomedical Applications (SMARTBIO-2025)

VIT-AP University, India

Received the **Best Oral Presentation Award** for presenting:

"Magneto-Optic Thermal Heating with Eu-Doped Fe₃O₄ Mesocrystals for Synergistic Cancer Therapy and Bioimaging."

This award recognized the scientific quality, originality, and presentation of my research.

SRM Institutional Fellowship

Recipient of the **SRM Institute of Science and Technology Institutional Fellowship** for pursuing doctoral research in the Department of Physics and Nanotechnology.

The fellowship supports my research on multifunctional magnetic mesocrystals for biomedical applications.

First Prize – IGNITE Science Exhibition (2021)

Received **First Prize** for presenting an innovative science project during the IGNITE Science Exhibition.

This recognition strengthened my interest in scientific research and motivated me to pursue a career in nanotechnology and materials science.

NEWS & UPDATES

2026

 Presented research at the **15th International Conference on the Scientific and Clinical Applications of Magnetic Carriers**, Sorbonne University, Paris, France.

 Received **ANRF International Travel Support** to present research in France.

 Presented **Eu³⁺-Doped Fe₃O₄ Mesocrystals for Magneto–Optic Synergistic Cancer Therapy and Bioimaging** at AFMD-2026.

2025

 Received the **Best Oral Presentation Award** at SMARTBIO-2025, VIT-AP University.

 Published my first research article in **Journal of Materials Chemistry B**.

 Published my first review article in **Journal of Solid State Chemistry**.

 Presented research at **ICANN 2025** (IIT Guwahati).

 Presented research at **ICONN 2025** (SRMIST).

2024

 Springer conference proceeding accepted for publication.

 Presented research at **IEMDST-2024**, NIT Warangal.

 Presented research at **iReAd MAGMA-2024**, VIT-AP University.

2023

 Joined the **Key Laboratory of Emergent Materials (KLEM)** as a Ph.D. Research Scholar.

Started research on **Magnetic Mesocrystals via Non-Classical Crystallization for Magnetic Hyperthermia Applications.**

TECHNICAL SKILLS

Nanomaterial Synthesis

- Hydrothermal Synthesis
 - Solvothermal Synthesis
 - Mesocrystal Engineering
 - Rare-Earth Doping
 - Surface Functionalization
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Characterization Techniques

- Powder X-Ray Diffraction (XRD)
 - Field Emission Scanning Electron Microscopy (FESEM)
 - Transmission Electron Microscopy (TEM)
 - High-Resolution TEM (HRTEM)
 - Fourier Transform Infrared Spectroscopy (FTIR)
 - UV–Visible Spectroscopy
 - Photoluminescence Spectroscopy
 - Vibrating Sample Magnetometry (VSM)
 - Dynamic Light Scattering (DLS)
 - Zeta Potential Analysis
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Biological & Biomedical Techniques

- Magnetic Hyperthermia Evaluation
 - Drug Loading and Release Studies
 - Cell Culture Handling
 - Cytotoxicity Analysis
 - Fluorescence Imaging
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Software & Data Analysis

- OriginPro
 - GSAS-II
 - FullProf Suite
 - ImageJ
 - COMSOL Multiphysics
 - XPS Peak4
 - Microsoft Office Suite
-

PROFESSIONAL MEMBERSHIPS

I actively participate in professional scientific communities to remain connected with current research developments and foster collaborations with researchers worldwide.

- American Physical Society (APS)
 - American Chemical Society (ACS)
 - ResearchGate
 - Google Scholar
 - ORCID
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LEADERSHIP & SERVICE

Scientific research requires more than technical expertise—it also demands teamwork, communication, and leadership. Throughout my academic journey, I have actively contributed to community initiatives and developed strong interpersonal skills.

Media Team Leader

Jesus With Us Church, Karungal, Tamil Nadu, India

Served as a **Media Team Leader** for three years, planning meetings coordinating, and mentoring team members. This experience strengthened my leadership, communication, decision-making, and organizational skills while teaching me how to effectively manage multiple responsibilities in collaborative environments.

BEYOND RESEARCH

I believe that maintaining a healthy balance between professional and personal life is essential for creativity, productivity, and continuous learning.

Sports

Sports have always been an important part of my life. I represented both my **college and club handball teams** for nearly **four years**, competing at the **state level**. These experiences taught me teamwork, resilience, discipline, and the importance of perseverance. Today, I continue to stay active by playing **badminton**, which helps me maintain physical fitness and mental well-being.

Photography & Media

Photography is one of my favorite creative pursuits. For more than **six years**, I have been involved in **photography, videography, and live media production** at **Jesus With Us Church, Karungal, Tamil Nadu, India**.

Through this experience, I have developed skills in visual storytelling, DSLR camera operation, live streaming, multimedia production, and event documentation. Photography has enhanced my creativity, attention to detail, and ability to observe the world from different perspectives.

Personal Interests

Outside of research, I enjoy:

- Watching Anime
- Travelling
- Exploring Emerging Technologies
- Reading Scientific Literature
- Photography
- Playing Badminton

I particularly enjoy anime because of its creativity, inspiring stories, and themes of determination, teamwork, and continuous self-improvement. These values often resonate with my own approach to research and personal development.

RESEARCH PHILOSOPHY

I believe that meaningful scientific research begins with curiosity and is driven by collaboration, creativity, and perseverance. My goal is not only to develop advanced materials with improved functional properties but also to translate these discoveries into practical biomedical technologies that benefit society.

I am passionate about interdisciplinary research, where materials science, physics, chemistry, and biology converge to solve complex healthcare challenges. By continuously learning new techniques, collaborating across disciplines, and embracing innovation, I hope to contribute to the advancement of precision medicine and inspire future researchers.

FUTURE GOALS

Looking ahead, I aspire to establish an independent research group dedicated to the development of multifunctional nanomaterials for biomedical applications.

My long-term research goals include:

- Designing next-generation magnetic mesocrystals for precision medicine.
- Developing multifunctional nanoplatfroms integrating diagnosis and therapy.
- Advancing fluorescence-guided cancer treatment.
- Exploring artificial intelligence and machine learning for materials discovery.
- Building national and international research collaborations.
- Mentoring young researchers and contributing to scientific education.

Ultimately, I hope my research will bridge the gap between fundamental materials science and clinical medicine, leading to technologies that improve patient care and quality of life.

CONTACT

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Research Profiles

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 - ResearchGate : [Akash Marsalin](#)
 - LinkedIn : [Akash Marsalin | LinkedIn](#)
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